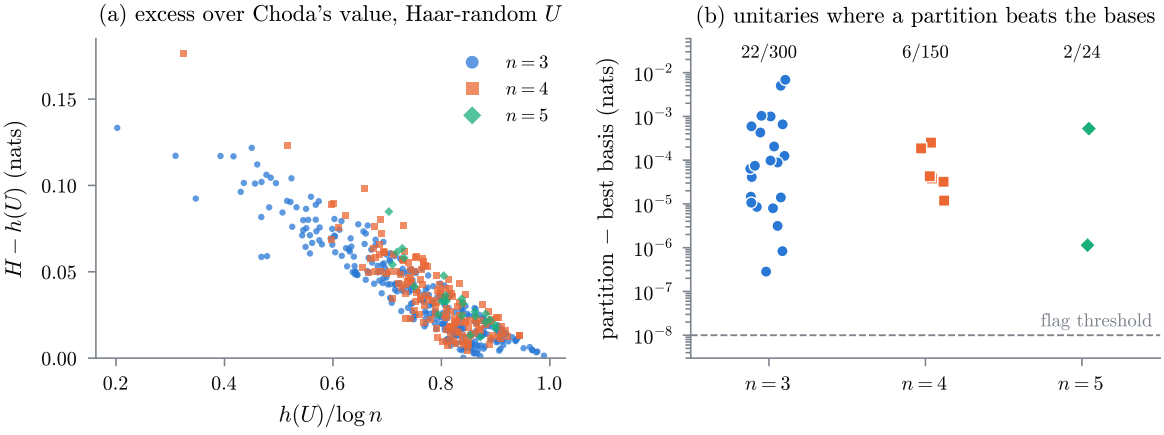
(a) excess over Choda's value, Haar-random U
$n = 3$
$n = 4$
$n = 5$
$H - h(U)$ (nats)
0.15
0.10
0.05
0.00
0.2
0.4
0.6
0.8
1.0
$h(U)/\log n$
(b) unitaries where a partition beats the bases
22/300
6/150
2/24
partition − best basis (nats)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
flag threshold
$n = 3$
$n = 4$
$n = 5$